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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
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Task Description:

Manual crew override during exploration spaceflight consists of both planned automated supervisory control and unplanned crew override. This crew override capability is critical to enable overall mission success during landing contingencies. However, the introduction of manual override capabilities must be implemented to enable crews to mitigate risks introduced by human error. Adaptive changes in the sensorimotor system can manifest during G-transitions as spatial disorientation. While training and landing aids enable successful landing through disorientation, these adaptive changes may increase cognitive demand that need to be accounted for in the manual control strategy. It is important to characterize these effects as soon as possible following the G-transition. The primary goals of this study are (1) to understand the impact of spaceflight on crew ability to perform manual crew override tasks including the effect of flight duration, (2) to examine how adaptive changes in vestibular and cognitive function relate to changes in manual crew override proficiency, and (3) compare performance during late in-flight “just-in-time” on-board training with early post-flight crew performance. This study will recruit astronauts assigned to either short duration Private Astronaut Missions (PAM, < 30 day) or long duration (6-month) missions to the International Space Station (ISS). Ground matched-control subjects tested with the same schedule will enable us to assess the effects of learning and/or recency independent of spaceflight.

Based on a thorough review of significant manual control events, it was concluded that at least 35 close calls could have resulted in loss of mission had crew manual control capabilities not been available to support continued mission operations (NASA Flight Safety Office 2013). While manual crew override has been critical to mission success (Johnson 2019), lessons learned include implementation strategies to minimize variability in performance. For example, while all six Apollo landings were successful, blowing dust obscured visibility during descent and appeared to contribute to variability in nulling horizontal rates, similar to brown-out disorientation during helicopter hover (Wagner 2006). Evaluation of Shuttle landings also suggested that performances during orbiter landings were more variable compared to the preflight Shuttle Training Aircraft landings. Key landing parameters such as sink rate and touchdown speed exceeded desired limits across several missions (Paloski et al. 2008; Moore et al. 2011). Data mining of the first 108 Shuttle landings showed 20% exceeded touchdown speeds and 12% exceeded touchdown sink rates (Moore et al. 2008; Clark et al. 2019). The results of some Shuttle experiments suggested adaptive changes in vestibular processing may have contributed to these post-landing manual control deficits (Merfeld 1996; Clément et al. 2018). Preliminary data mining of the postflight neurological exams suggested impaired dynamic equilibrium was associated with Shuttle landings that were shorter, faster, and harder (McCluskey et al. 2001). Our first aim is to characterize the impact of spaceflight on crew capability to perform automated supervisory control and manual landing tasks involving crew override following both short (PAM) and longer duration ISS missions. The impact of spaceflight on piloting capability will be assessed from pre- versus post-flight changes in a simulated lunar landing during which crewmembers will manually takeover attitude and rate-of-descent to the nominal or re-designated landing aimpoint during the approach phase. This simulation will be performed on portable limited 3DOF (degree-of-freedom) motion base soon after return from ISS at the rally airport and then on R+1 using a 6DOF motion base at JSC. Subjects will also perform an automated supervisory control on a portable tablet (named ATLAS). During this task, subjects will monitor simulation of an out-the-window display during lunar approach to determine if they need to execute a divert to an alternate landing site and use a touch screen to select the site that is within the desired landing zone and free of obstacles. The outcome measures during the piloting simulations include the ability to maintain actual vehicle states, e.g., pitch, roll, heading and sink rate errors relative to recommended guidance during the landing, and errors relative to the landing target, and subjective workload measures. The supervisory task outcome measures include time and altitude to select divert, hazards and ground slope at new landing target, and whether the divert was within the vehicle’s capabilities. The cognitive performance measures during the simulations include reaction time to respond to display prompts as a secondary task and number of indicators missed, as well as eye-head tracking to characterize visual reaction times and areas of visual attention during the simulation. We hypothesize that there will be post-flight increases in error rates and time to execute corrective actions.

Our second aim is to examine how adaptive changes in vestibular and cognitive function relate to proficiency during supervisory manual control and crew override. The risk of spatial disorientation leading to aircraft mishaps is multi-faceted involving visual, vestibular, somatosensory, and psychological mechanisms among other factors (Previc and Ercoline 2004; Parmet et al. 2021). This test battery is based on a previous manual control study by Moore et al (2019) in which astronauts exhibited significant changes in dual-tasking, motion perception, and sleepiness. As described above, the dual-tasking has been embedded in both tablet and piloting tasks. Motion sickness, sleepiness and fatigue subjective scales will be obtained each session. Motion perception has been integrated with the motion-base sim at JSC using a closed-loop nulling task. During this task, nulling performance, e.g., error rates and lag times, will be obtained during closed-loop nulling of pseudorandom roll-tilt disturbances while viewing a lunar terrain (control condition), with eyes closed (vestibular and proprioceptive cues alone), and during uncorrelated visual motion that represents brown-out type conditions. We hypothesize that that a higher severity of vestibular alterations will be associated with increased flight increases in error rates and time to execute corrective actions.

Decrements in dual-tasking may be related to alterations in time perception during G-transitions (Clément 2018) during which “time compression” may result from either loss of task proficiency or lack of cognitive reserve. The "Portable In-flight Landing Operations Trainer" (PILOT) was a laptop computer simulator that was flown as a tool for helping the mission commander and pilot maintain their proficiency for approach and landing during longer duration Space Shuttle flights (Kennedy et al. 1997; Life Science Data Archive 2020). One Shuttle commander who experienced vertigo during the landing (personal communication) attributed the “just-in-time” training of the PILOT to be able to lower perceived cognitive workload and effectively pilot through disorientation to complete a successful landing. The success of the Shuttle PILOT serves as a basis for our proposed “just-in-time” training countermeasure to mitigate risks associated with spatial disorientation during lunar landings by maintaining peak crew task training proficiency. While the fidelity of the inflight laptop trainers is limited, for the purpose of maintaining task proficiency the Shuttle PILOT was highly effective.

Our third specific aim is to compare performance during “just-in-time” onboard training on the lunar landing tasks conducted late in-flight with post-flight crew performance. Given that ‘just-in-time” on-board training is an operational expectation for the extended exploration missions, all participants will perform late inflight on-board training. For our implementation of on-board training for the lunar lander simulation, we plan to leverage an existing platform already on the ISS referred to as the Robotic On-Board Trainer developed for onboard training on the Canadarm2 track-and-capture activities (ROBoT, Ivkovic et al. 2019). The metrics captured during the on-board lunar landing simulations will match those of the pre- and postflight simulations. We hypothesize that that proficiency on the “just-in-time” laptop trainer late in mission will be positively correlated with early post-flight proficiency on the same task.

This project will deliver an operational demonstration of automated supervisory control and crew override capability

	following spaceflight and identify potential deficits that may require remediation. Comparison of individual vestibular and cognitive changes with crew override performance will help better characterize the manual control risks associated with sensorimotor alterations. Comparing performance parameters from the on-board training to post-flight performance will help demonstrate transfer of inflight on-board training to post-flight manual crew override performance. The inclusion of “just-in-time” on-board training will ensure we are characterizing changes in override proficiency with this expected countermeasure in place.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This flight study addresses the sensorimotor research emphasis stated in the Human Research Program (HRP) Integrated Research Plan titled “Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks”. One gap associated with this risk (SM-102) is to “characterize the effects of short and long-duration weightlessness on manual control after G-transitions.” This research gap led to the solicitation of a manual control study conducted before and after long duration flights on the ISS to map changes in sensorimotor function to manual control decrements. Unfortunately, these results (Moore et al., 2019) were limited due to testing delays related to time required for direct returns from Kazakhstan, with the initial measurements conducted more than 20 hr following landing. The approach of this investigation is to leverage the commercial crew landings in the US to obtain measurements as early as possible. While the Moore study used a T-38 X-plane simulation, we will obtain measures during actual T-38 flights. We will also add a lunar landing simulation based on the current concept of operations for the HLS. Further refinements are also proposed in the sensorimotor and cognitive test battery based on the previous study.
Research Impact/Earth Benefits:	Sensorimotor function is critical for spatial orientation, gaze stabilization, and postural stability. This project examines how adaptive changes in sensorimotor and cognitive function may increase the risk of impaired ability to maintain control of vehicles and other complex systems. The goal is to map changes in physiological function with functional measures of manual control. Establishing these relationships will be relevant to how pathophysiological impairments in sensorimotor processing may affect other vehicular control tasks, such as driving with vestibular patients. This study will also be relevant to the development of strategies of how best to allow individuals with impairments perform a manual control override of autonomous driving vehicles. As Shuttle missions were extended, the on-orbit training countermeasure tool became extremely useful for commanders and pilots to practice the landing task sequence to maintain task proficiency. The ability to retain and/or assess manual control efficiency via a simple laptop-based simulation will impact our crew override approach for extended exploration missions. This approach can be easily adapted to a wide variety of simulated vehicle designs to provide similar assessments in other operational and civilian populations.
Task Progress:	The accomplishments are described below for three main areas: (a) hardware and software integration, (b) the International Space Station (ISS) flight study, and (3) the supporting ground studies. 2a. Hardware and Software Integration The project team, comprised of the investigators, the Charles Draper Laboratory, and the Dynamic Skills Training Lab, supported rigorous development and testing of the Lunar Landing Simulation (LLSim) application integrated with the Robotic On-board Training (ROBoT) platform across three unique system environments. The three target environments included the Neuroscience laboratory with a 6DOF motion base, a portable simulator for use in post-flight field testing, and finally the ROBoT platform in-flight on the ISS (Figure 1). The team also collaborated to support LLSim and ROBoT operations across three laptop configurations (G2, G7, and G9), one workstation (Titan), and two operating systems (Oracle Linux 7 and Oracle Linux 8), resulting in seven unique configurations. Each configuration and target environment presented its own set of challenges. Ultimately, the team developed workarounds, patches, and targeted fixes to ensure all seven configurations performed as intended. The software review and approval from Marshall’s Payload Operations Integration Center (POIC) was completed in February 2025. Notably, the team accomplished the ground certification of the LLSim using the Joint System Local Area Network (JSL) laboratory in June 2025. Uplink to station was completed in August 2025 with support from the EVA, Robotics and Crew Systems Division (CX). The Research Operation and Integrations team facilitated an in-flight checkout of the hardware and software on station with the first consent crew member for the Manual Piloting Flight study. This in-flight checkout was necessary because it uncovered critical failures in the data server path, along with inconsistencies in the landing trial scenarios. These issues were promptly resolved, and a second in-flight checkout was performed a month later in October 2025. The second checkout verified readiness for the first in-flight data collection. A concerted effort was also performed by the Neuroscience team and the Simulation and Graphics Branch of the Engineering Directorate (ER7) to complete the development of an iOS-based multi-attribute supervisory control lunar tablet task. Completion of the tablet task was achieved in February 2025. The application was on its eighteenth iteration when it was transferred to the Neuroscience Laboratory for maintenance. Over the course of the year, the Neuroscience team completed ten additional iterations, improving usability, strengthening data verification, and implementing key updates based on feedback from ground testing and astronaut participants. The ER7 team delivered the final post-processing tool in May 2025. The Neuroscience team accomplished the software development of a closed-loop roll nulling task with and without visual disruption in May 2025. The application was developed as the sensorimotor test battery for the Manual Piloting flight study’s preflight and postflight testing sessions, capturing critical data to address the study’s second specific aim: examining how adaptive changes in vestibular and cognitive function relate to proficiency during supervisory manual control and crew override. The Neuroscience team recently purchased a 3DOF motion system to serve as an improved portable simulator and enable higher-quality data collection at the FBO testing sites on landing day. The system provides limited motion capability. The team also completed computer-aided design for a custom build incorporating a simulator seat, monitor array, and hand controllers. A stress analysis and tip analysis were completed in January 2026. 2b. ISS Flight Study HRMRB, ESA, and JAXA medical board approvals were obtained by February 2025. The first consented crew member completed the research protocol in January 2026. Although not all sessions were completed due to mission-related circumstances, sufficient data were collected, including the earliest feasible postflight session conducted one day following landing (R+1). Two additional crew consents were obtained from Crew 12, and all preflight training and

	baseline data collections were completed from October 2025 through January 2026. Informed crew briefings (ICB) were provided to Crew 13, and two participants are anticipated. The Crew 14 ICBs are targeted for May 2026. As mentioned in 2a, all four prime Artemis II crew and two backups consented to perform the closed-loop nulling task once preflight and early postflight upon return to JSC. All preflight data collection was completed, and the postflight data is pending the successful completion of the flight, which is targeted for no earlier than March 2026. 2c. Supporting Ground Studies The Neuroscience Laboratory team completed the data collection of the Manual Piloting Ground Learning Study. The study involved thirty healthy non-astronaut volunteers who each completed five testing sessions, totaling 150 hours of simulated lunar landing trials on a 6DOF motion base, along with a custom iOS-based supervisory control tablet task from January 2025 through July 2025. In total, 1,800 manual control landing scenarios and 720 supervisory landing scenarios were conducted. Data collection focused on pilot performance, gaze tracking, workload, and skills retention following a four-month hiatus using the system. The findings from this work provide insight into the time required for both non-pilot and pilot individuals to achieve proficiency on the landing task, as well as the retention of these skills after a four-month period without practice. These findings directly support the Manual Piloting ISS Flight Study by characterizing user performance under nominal sensorimotor conditions to establish baseline proficiency and to refine the training objective. An abstract summarizing the results of this ground study was submitted to the 2026 NASA Human Research Program Investigators' Workshop and will be presented in April 2026. In January 2026, the research team shared preliminary results with the Flight Operations Directorate Community comprised of the Crew Training Officers, to inform the use of objective measures during foundational skills training.
Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Weiss HM, Widhalm D, Bollinger AM, Duda KR, Wheelock DH, Wood SJ. "Proficiency training and retention for manual crew override during simulated lunar landing." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026
Articles in Peer-reviewed Journals	Allred AR, Lippert AF, Wood SJ. "Galvanic vestibular stimulation advancements for spatial disorientation training." Aerosp Med Hum Perform. 2024 Jul;95(7):390-8. https://doi.org/10.3357/AMHP.6362.2024 ; PubMed PMID: 38915170 , Jul-2024